

JOSEPH RIVAS

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Senior Machine Learning Engineer with expertise in designing and deploying LLM-powered and trustworthy AI solutions across healthcare, government, and enterprise domains. Skilled in retrieval-augmented generation (RAG), domain adaptation, and MLOps, with a track record of building scalable, production-ready AI systems that prioritize reliability, compliance, and transparency. Strong foundation in biomedical research, aerospace, and engineering, complemented by advanced training in computer science and applied machine learning.

EXPERIENCE

Machine Learning Engineer | Jaxon.ai | Boston, Massachusetts

February 2025 – Present

- Building and deploying **trustworthy AI systems** leveraging Jaxon's proprietary **DSAIL (Domain-Specific AI Language)** technology, which applies formal constraints and assertions to validate model outputs.
- Conducting **LLM research and engineering** to reduce hallucinations and ensure reliability in mission-critical applications for government and enterprise clients.
- Designing **retrieval-augmented generation (RAG) pipelines** and domain adaptation strategies to improve contextual accuracy and grounded responses.
- Collaborating with cross-functional teams to integrate **responsible AI practices**, ensuring compliance, transparency, and verifiability in production systems.
- Prototyping and testing **constraint-driven LLM architectures**, enhancing safety, robustness, and explainability of AI-driven decision support.
- Leveraging cloud infrastructure, vector databases, and MLOps frameworks to operationalize advanced LLM solutions at scale.

Machine Learning Engineer | The Janssen Pharmaceutical Companies of Johnson & Johnson | Titusville, New Jersey

August 2023 – September 2024

- Designed and deployed **LLM-powered solutions** to accelerate scientific knowledge extraction, including summarizing clinical trial results and generating research insights from biomedical literature.
- Implemented **retrieval-augmented generation (RAG) pipelines** to enable domain-specific question answering over proprietary datasets, improving knowledge accessibility for researchers and clinicians.
- Developed and fine-tuned **large language models (OpenAI, Hugging Face, LLaMA-based architectures)** for biomedical text understanding and patient-centric applications.
- Built scalable **data ingestion and retrieval systems** (vector databases, embeddings) to support LLM-driven workflows across R&D.
- Applied **MLOps best practices** for LLM lifecycle management, including prompt engineering, evaluation, and continuous improvement.
- Collaborated with cross-functional teams to integrate **LLM applications** into clinical research and decision support tools, enhancing productivity and accelerating discovery.

Machine Learning Engineer | Lightspeed Systems | Austin, TX

July 2021 – December 2022

- Designed, implemented, and maintained **custom MLOps infrastructure** to streamline model development, training, deployment, and monitoring across production environments.
- Automated **end-to-end ML pipelines** (data preprocessing, model training, validation, and deployment) improving efficiency and reducing time-to-production.

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- Developed APIs and scalable deployment strategies (Docker, Kubernetes, CI/CD) for serving machine learning models with high availability and reliability.
 - Optimized ML workflows by introducing **version control, experiment tracking, and reproducibility** using tools such as MLflow, DVC, and Git.
 - Collaborated with product and data science teams to align **ML model development with business requirements**, ensuring scalability, security, and compliance.
 - Enhanced system performance through continuous monitoring, alerting, and retraining mechanisms, ensuring models remained accurate and robust over time.

AI/ML Engineer | Lockheed Martin | Santa Clarita, CA

June 2019 – July 2021

- Developed and deployed **machine learning and computer vision models** using Python (NumPy, Pandas, scikit-learn, TensorFlow, Keras, OpenCV) to support aerospace and defense applications.
- Built scalable **data pipelines and workflow automation** with Luigi, Bash, and Docker, improving efficiency of model training and deployment.
- Designed and maintained **interactive web applications and dashboards** with Flask, Angular 9, JavaScript, and Plotly for data visualization and decision support.
- Collaborated with cross-functional teams to integrate **AI/ML capabilities** into production systems while ensuring security, performance, and compliance.
- Applied best practices in **MLOps**, containerization, and CI/CD to deliver reliable, production-grade solutions.
- Contributed to projects in **data analytics and predictive modeling**, enabling data-driven insights for business and engineering stakeholders.

R&D Engineer | Medtronic | North Haven, CT

September 2017 – March 2019

- Designed and automated mechanical, electrical, and software test methods, generating large datasets to evaluate surgical robotics system performance.
- Applied statistical modeling and Gage R&R analyses in **Minitab** to validate measurement protocols, ensuring reliability and reproducibility of data-driven insights.
- Developed reliability validation workflows (IQ/PP/AP) that parallel modern AI/ML model validation and system testing practices.
- Contributed to risk assessments (DFMEA) and human factors usability studies, aligning system testing with predictive analytics for failure detection.
- Leveraged **PLM and project tracking tools (JIRA, Jama, Oracle/Agile)** to manage system requirements, testing data, and compliance workflows across interdisciplinary teams.

R&D Engineer | BD | Franklin Lakes, NJ

May 2016 – April 2017

- Designed and automated mechanical, electrical, and software test methods, producing datasets to evaluate surgical robotics and medical device system performance.
- Applied statistical modeling and data analysis techniques (CPK, Gage R&R, ANOVA, reliability studies) in **Minitab**, strengthening foundations in data-driven experimentation relevant to AI/ML pipelines.
- Conducted root-cause investigations and failure analyses, employing structured problem-solving approaches similar to modern anomaly detection and model debugging.
- Developed and validated test fixtures, protocols, and measurement systems, ensuring data quality, reproducibility, and compliance with FDA (510k) and ISO 13485 standards.
- Collaborated cross-functionally on risk assessments (DFMEA), usability studies, and regulatory submissions, highlighting skills in governance and responsible system development.
- Leveraged PLM and project management tools (JIRA, Jama, Oracle/Agile) to track requirements, testing progress, and compliance workflows across interdisciplinary teams.

Mechanical Engineer | Amphenol | St. Marys, PA

August 2015 – May 2016

- Used Pro E, Solidworks, and AutoCAD software to evaluate, design and develop products compliant with ANSI/ISO and ASME Y14.5 standards, engineering principals, and customer contract specifications; created models and technical prints
- Employed rapid prototyping methods using 3D Systems ProJet printer in order to expedite design process
- Successfully led design efforts and coordinated with manufacturing, quality, and engineering groups to introduce products and implement engineering change notices (ECN) suitable to customers' requirements
- Applied finite element analysis (ANSYS) to model structural performance, an early foundation in simulation and predictive modeling methods relevant to AI/ML.
- Enhanced project management skills by scheduling customer meetings, contacting vendors for quotes, calculating price adjustments, and ensuring timely work completion by assigned engineering group
- Successfully employed Spanish language skills to facilitate communication with Amphenol production team in Tijuana, Mexico

Undergraduate Researcher | University of Michigan | Ann Arbor, MI

June 2013 – August 2013

- Independently designed experiments for characterization and deposition of magnetic permalloy thin films for future endomicroscope actuator applications, emphasizing reproducibility and data-driven analysis.
- Learned SEM/EDX techniques to conduct chemical composition analyses, conducted metrology experiments using stylus/laser profilometers; used variety of microfabrication techniques including evaporation and sputtering, PECVD, and electroplating to apply thin films to silicon wafers; became proficient in variety of cleanroom tools and processes
- Strengthened ability to work both independently and in groups; greatly improved presentation skills for final research exposition held at Georgia Institute of Technology

EDUCATION

Bachelor of Science in Mechanical Engineering | Binghamton University

Degree obtained June 2011

Master of Science in Computer Science | Georgia Institute of Technology

Degree obtained June 2013